

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AAE5110
Subject Title	Air Transport Economics and Policy
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject will provide students with 1. an advanced understanding of the economic principles and theories that underpin the air transport industry and its regulation; 2. a critical analysis of the current issues and challenges facing the air transport sector, such as market structure, competition, pricing, demand, supply, externalities, social costs and benefits, and sustainability; 3. an appreciation of the role and impact of various policies and institutions that shape the air transport environment, such as international agreements, and environmental regulation; and 4. an opportunity to apply economic concepts and tools to evaluate and propose solutions for real-world problems and cases in the air transport industry.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. understand economic principles and concepts relevant to air transport. b. analyze the economic impact of air transport on economies and interconnected sectors. c. evaluate and propose improvements to air transport policy frameworks. d. apply economic and policy analysis to address industry challenges and propose evidence-based recommendations for performance and sustainability.
Subject Synopsis/ Indicative Syllabus	Introduction to Air Transport Economics: The role and importance of air transport in the global economy; key economic concepts such as opportunity cost, the value of money, supply and demand, imperfect markets, market systems (monopoly, oligopoly), and quantity control; the characteristics and structure of the air transport market; the economic models and tools for analysing the air transport market. Air Transport Cost and Production: The components and determinants of air transport costs; the economies of scale and scope in air transport production; the cost and production functions and their properties; the

	productivity and efficiency measures and indicators for air transport; resource allocation for cost reduction and service improvement. Air Transport Market Structure and Competition: The market structure and conduct of the air transport industry; the impacts of market power and competition on air transport performance and welfare; the antitrust and competition policies and regulations for the air transport industry. Air Transport Policy and Regulation: The objectives and instruments of air transport policy and regulation; the evolution and development of air transport policy and regulation; liberalisation in aviation market; the impacts and implications of air transport policy and regulation on air transport performance and welfare. Air Transport Pricing and Revenue Management: The pricing strategies and methods in the air transport industry; the revenue management and optimization models and practices; the price competition among airlines.																			
Teaching/Learning Methodology	Teaching is conducted through class lectures and tutorials. Lectures are aimed at delivering the core knowledge and concepts of air transport economics and policy and policy regulations relevant to air transport. Tutorials are used to reinforce the understanding and application of the economic concepts and tools through exercises and discussions. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	√	√	√	√	Tutorial	√	√	√	√
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	a	b	c	d																
Lecture	√	√	√	√																
Tutorial	√	√	√	√																

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1. Assignment	30%	√			√
	2. Group project	20%	√	√	√	√
	3. Final examination	50%	√	√	√	√
	Total	100 %				
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 0.5 × Continuous Assessment + 0.5 × Final Examination The continuous assessment (50%) is designed to enhance the students’ comprehension and assimilation of various topics of the syllabus through assignments and projects. In particular, the assignments will evaluate students' understanding of economic principles and concepts relevant to air transport, as well as their ability to apply economic and policy analysis to address challenges in this field. Additionally, the group projects will assess students' capacity to apply economic and policy analysis to solve real-world cases. The final examination (50%) will also evaluate the students' learning outcomes. The final examination will test the students' comprehension of economic and policy analysis related to air transport.					
Student Study Effort Expected	Class contact:					
	▪ Lecture					39 Hrs.
	Other student study effort:					
	▪ Self-learning/preparation					36 Hrs.
	▪ Literature study/case study/reading					36 Hrs.
	Total student study effort					111 Hrs.
Reading List and References	1. Mankiw, N. Gregory. Principles of economics. Cengage Learning, 2020. 2. Mankiw, N. Gregory, Ronald D. Kneebone, Kenneth J McKenzie. Principles of Macroeconomics, Cengage Canada, 2023. 3. Holloway, Stephen. Straight and level: Practical airline economics. Ashgate Publishing, Ltd., 2008. 4. Morrell, Peter S. Airline finance. Routledge, 2021. 5. Vasigh, Bijan, Ken Fleming, and Thomas Tacker. Introduction to air transport economics: from theory to applications. Routledge, 2018. 6. Graham, Anne, et al., eds. Air Transport and Regional Development Policies. Routledge, 2020.					